

Garden Clicker

By: Kiara Bartuccio and Oliver D'avino

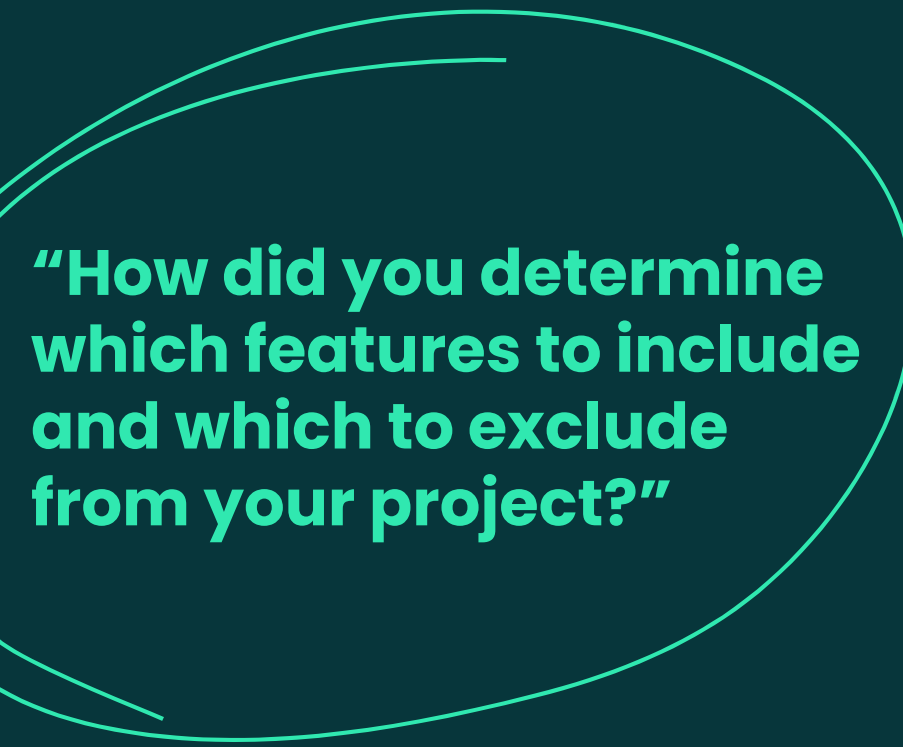
Two curved teal lines, one above the other, positioned below the authors' names.

About Garden Clicker



- 2D Pixelated clicker game
- related to Cookie Clicker
- Click the flower to increase flower count
- Buy upgrades to passively increase flowers and grow your garden
- Progress through rebirth mechanics
- Compete on leaderboards

Our objective was to design and develop an engaging mobile game that is easy for all users to navigate, relaxing/low-stress game, include goals that feel rewarding, and to keep the players engaged.



“How did you determine which features to include and which to exclude from your project?”

We analyzed the requirements needed for this project. We knew we wanted to do something different than others so we decided on a game. We came across the game “Cookie Clicker” and that was our main source of inspiration. Then we brainstormed together and combined our interests and put our own spin to the game.

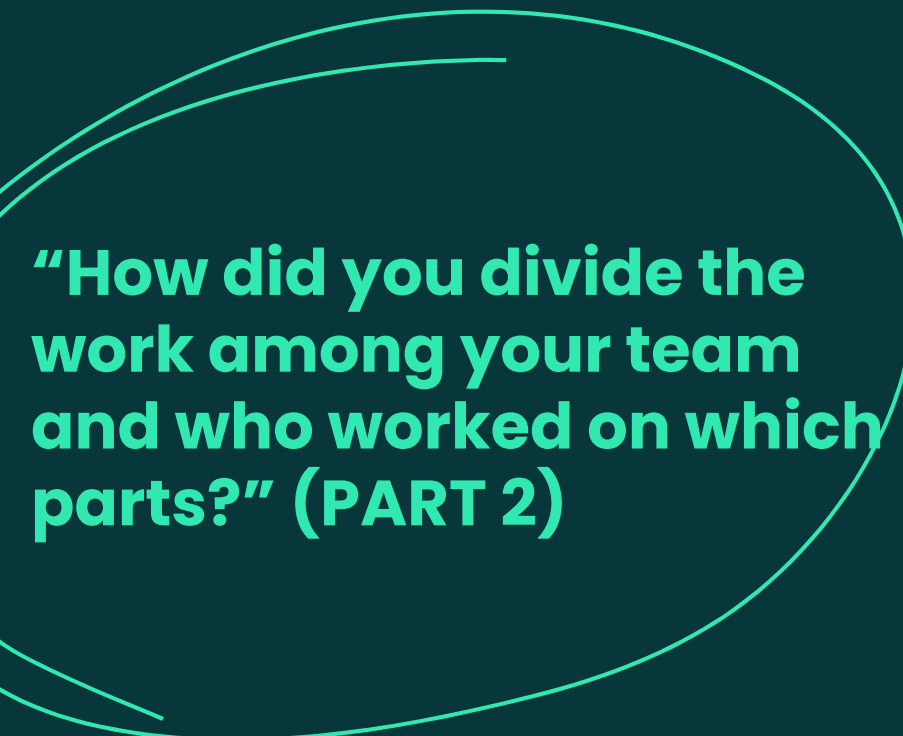
"How did you divide the work among your team and who worked on which parts?" (PART 1)

We tried to divide the work based on each others strengths:
Kiara(Firebase, navigation flow between screens, UI components like buttons)

- Firebase
- Splash screen
- Achievements(Expensive/Inexpensive)
- Main Menu:
 - Login
 - Play
 - Quit
- Settings
- Authentication with forgot password (EmailUS, ets the app send emails from the frontend without our own backend server)
- Documentation(Both P.P and the report)
- Exit

Oliver(Model classes, firebase, sprites, widgets, achievement + upgrades)

- Game save Firebase
- Art Direction
- Upgrade definitions (normal + rebirth)
- Leaderboard + Profile
- Main Game Screen:
 - Upgrade GridView
 - Rebirth Tab
 - Achievement Tab



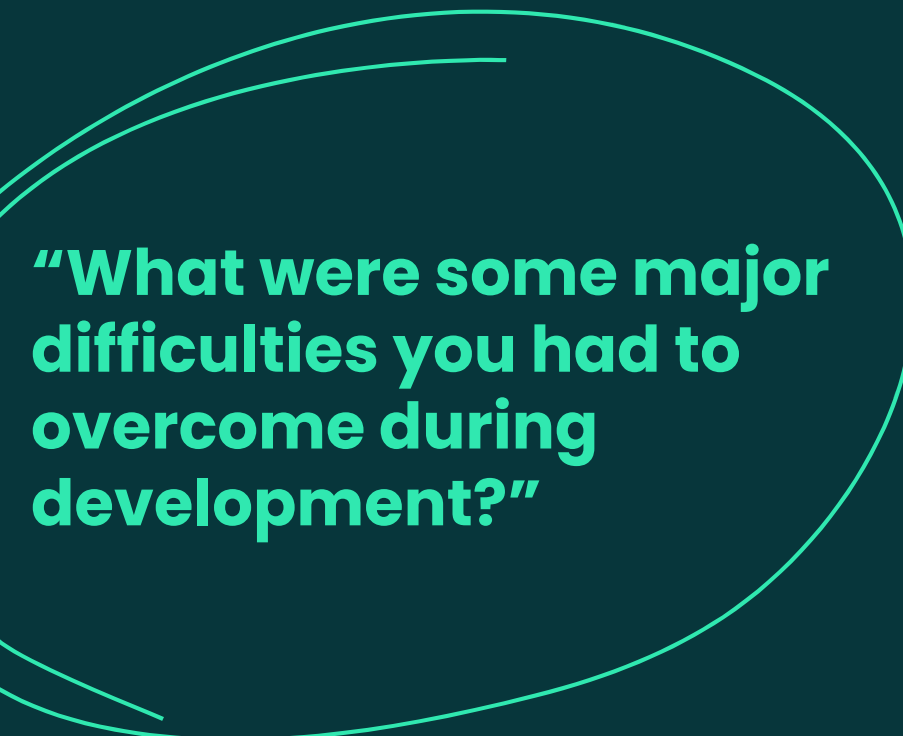
“How did you divide the work among your team and who worked on which parts?” (PART 2)

Shared Responsibilities

Parts of the project were also done together as it was just easier to complete:

- Thinking of the whole project idea
- What features we wanted to implement
- Designing the figma mockup
- Testing and debugging along the way (any problems we fixed them together)

Although we had to divide certain parts, most of the time we shared each task depending if the other person got stuck/had a problem or just did not have enough time to finish a part they were working on.



“What were some major difficulties you had to overcome during development?”

- **Game Development:**

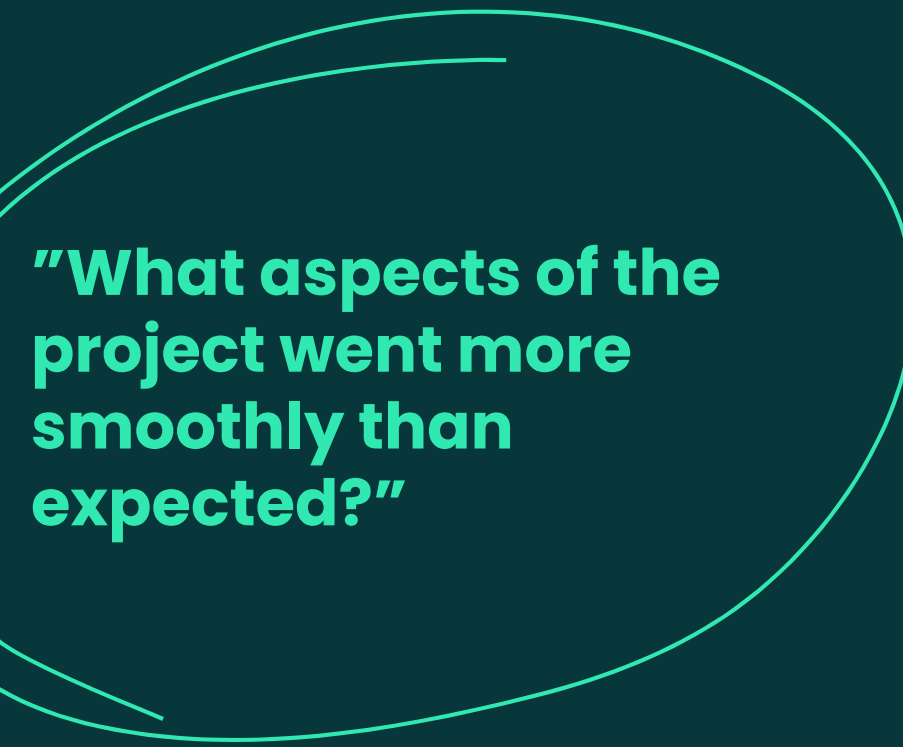
We do not build games as much as we do websites or other programs, so a game was a little bit of a challenge to take on since it was different than what we were used to. We had to do a lot of research to understand Flame and Firebase

- **Github:**

We had many issues with Github. We attempted to make different branches to allow both of us to work on the project simultaneously, however, it did not work and broke our project. It set us behind and wasted a lot of time trying to fix the error.

- **Time Management:**

It was difficult trying to get all the deliverables done in time, especially deliverable 2, since we had final exams and other final projects to complete all at the same time so it set us back a little.



"What aspects of the project went more smoothly than expected?"

- Screen Routing:

Our application included multiple screens, like splash screen, login, main menu, settings, rebirth screen, and more, we expected it to be a lot harder and more complicated than it was. Flutter's navigation system made it easy to transition between the screens once we set out plan into place.

- Upgrade Manager:

Implementing a system that lets players buy upgrades and build their garden was easier than expected. We found it easier to organize once we decided on a set plan. Our structure made it simple to keep track of the upgrade cost, increases, and the dynamic display.



"How can you improve your project in the future?"

- Add more functionalities

We could always expand the game by adding more upgrades, items to unlock, special events, daily login bonuses, player customization..

- Improve Visual Design:

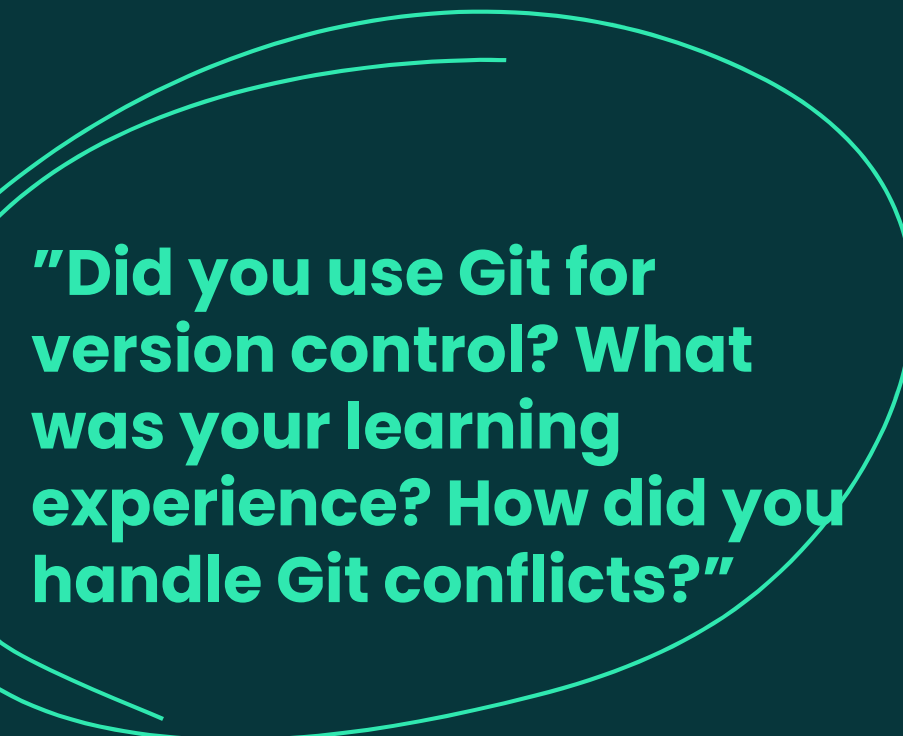
There is always room for improvement and there are always details to add to the garden and overall display of our game.

- Mini games

Mini games can be a game mode where the players can earn bonus flowers and special rewards. This will give the users another options beyond clicking the flower.

- Different worlds

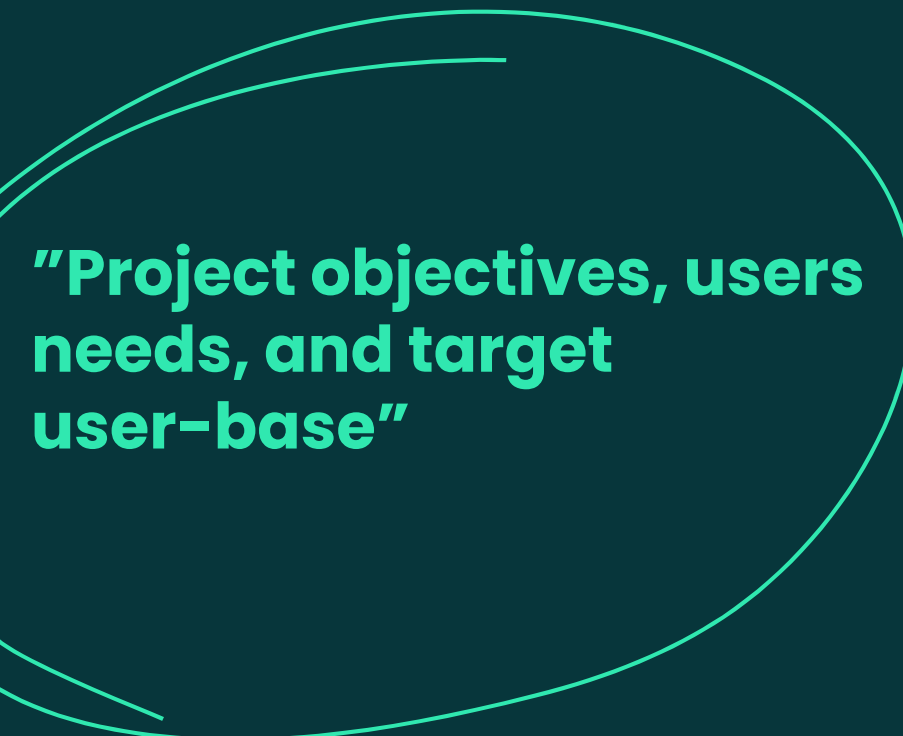
Once the user fully builds the garden, there can be a different world to build the garden in



"Did you use Git for version control? What was your learning experience? How did you handle Git conflicts?"

As mentioned before, we had many issues with Github. We tried to make different branches to allow both of us to work on the project at the same time, however, it did not work and broke our project. It set us behind and wasted a lot of time trying to fix the error. So in the end, we just would just work on it together in class on one computer, or wait for the other to be done, pulled, and worked.

We learned a lot, especially having to deal with that issue. We expanded our knowledge on cloning, committing, pushing, pulling, and understanding the history.



"Project objectives, users needs, and target user-base"

Main objective

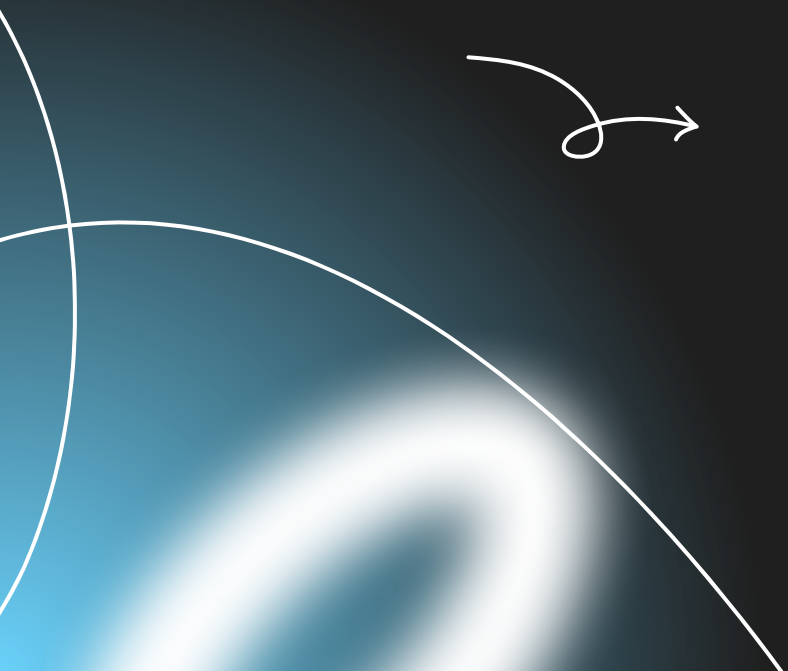
- Develop an engaging and enjoyable game that is: fun, relaxed, simple, and rewarding.
- We wanted the game easy to understand how to play and what the goal was so users do not lose interest or get frustrated.

User's needs

- Quick entertainment
- A way to decompress
- Low-stress
- Good for people who have short amounts of free time
- Passive progression
- Rewarding system

Target Users

- Casual gamers
- Fans of clicker games
- Teenagers, students., kids, adults
- People who are looking for a lightweight fun game that is played at their pace.



Thank You!
